

VEGREVILLE, AB, Canada

WMO: 714580

Lat: 53.5053N

Lon: 112.0978W

Elev: 2097

StdP: 13.62

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.7	-22.9	-36.9	0.7	-28.6	-30.7	1.1	-22.8	27.2	26.1	24.2	22.3	5.4	170	0.701

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.2	83.3	63.0	80.1	61.9	77.0	60.6	66.6	78.1	64.6	75.6	62.8	73.0	9.7	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.7	92.1	70.7	60.6	85.2	69.4	58.6	79.5	66.9	32.4	78.3	30.8	75.7	29.4	72.9	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.9	20.7	18.1	DB	-36.8	88.9	7.8	3.0	-42.4	91.0	-47.0	92.7	-51.4	94.4	-57.1	96.6	(4)
			WB	-37.1	70.6	7.6	2.4	-42.6	72.3	-47.0	73.7	-51.3	75.1	-56.8	76.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	36.0	8.0	10.7	21.2	38.2	50.6	58.1	62.4	60.0	51.1	38.3	21.8	10.3	(6)
(7)		DBStd	22.59	16.15	13.84	14.22	9.80	8.08	5.69	5.12	5.83	7.69	8.43	12.65	14.65	(7)
(8)		HDD50	6287	1301	1100	893	366	91	7	0	4	77	371	847	1230	(8)
(9)		HDD65	10638	1766	1520	1358	805	449	217	111	173	418	829	1297	1695	(9)
(10)		CDD50	1186	0	0	0	11	110	250	385	313	110	7	0	0	(10)
(11)		CDD65	63	0	0	0	0	3	9	32	17	2	0	0	0	(11)
(12)		CDH74	1397	0	0	0	5	128	202	504	409	146	3	0	0	(12)
(13)		CDH80	294	0	0	0	0	24	34	119	83	34	0	0	0	(13)
(14)	Wind	WSAvg	8.3	8.0	7.8	8.4	9.5	9.5	8.7	7.8	7.3	8.1	8.6	7.8	7.8	(14)
(15)	Precipitation	PrecAvg	16.3	0.7	0.4	0.7	1.2	1.7	2.8	3.2	2.1	1.3	0.7	0.7	0.6	(15)
(16)		PrecMax	22.6	2.7	0.9	1.5	2.6	4.6	5.0	5.5	4.1	3.2	1.7	3.0	1.5	(16)
(17)		PrecMin	8.7	0.1	0.0	0.2	0.1	0.3	0.8	1.2	0.4	0.2	0.0	0.0	0.1	(17)
(18)		PrecStd	2.9	0.6	0.2	0.3	0.6	0.9	1.1	1.1	0.7	0.8	0.4	0.6	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.6	41.3	52.4	73.0	83.4	84.0	88.2	86.6	84.8	70.5	53.5	44.1	(19)
(20)		MCWB	35.7	35.7	42.4	51.9	57.4	62.2	66.8	64.8	60.5	53.6	43.9	36.1	(20)	
(21)		2%	DB	38.7	37.5	45.6	66.1	77.4	79.4	83.0	82.2	78.1	62.7	45.7	39.4	(21)
(22)		MCWB	34.3	33.2	38.2	48.8	55.1	59.6	65.5	63.8	58.2	49.5	38.8	34.3	(22)	
(23)		5%	DB	35.2	34.3	41.1	60.0	72.7	75.6	79.5	78.6	72.9	57.5	41.3	35.2	(23)
(24)		MCWB	31.9	31.0	35.2	45.8	53.4	58.5	63.9	62.7	56.6	46.3	35.9	31.3	(24)	
(25)		10%	DB	30.7	30.7	37.9	54.9	68.2	72.1	76.1	75.2	67.4	52.5	37.5	31.0	(25)
(26)		MCWB	27.9	27.7	33.7	43.2	51.1	57.0	62.3	61.4	54.0	43.5	33.8	28.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.7	36.2	43.3	53.1	59.7	65.6	70.1	69.0	61.9	54.3	43.8	36.8	(27)
(28)		MCDWB	40.8	40.0	51.1	70.2	75.8	78.6	81.9	79.5	77.6	68.5	52.0	41.9	(28)	
(29)		2%	WB	34.5	33.5	38.9	49.4	56.7	62.6	67.4	66.2	59.4	50.4	39.6	34.1	(29)
(30)		MCDWB	38.2	36.8	44.6	63.6	72.7	73.6	79.2	77.6	75.2	59.7	44.6	38.4	(30)	
(31)		5%	WB	32.0	30.9	36.0	46.4	54.6	60.6	65.4	64.2	57.2	47.4	36.5	31.3	(31)
(32)		MCDWB	35.6	34.2	40.5	58.5	69.2	70.9	76.3	75.2	71.3	55.8	40.5	34.9	(32)	
(33)		10%	WB	27.9	27.7	34.0	43.6	52.5	58.6	63.6	62.2	54.9	44.4	34.0	27.7	(33)
(34)		MCDWB	30.7	30.5	37.8	53.3	66.0	68.8	73.5	72.7	65.8	51.6	37.2	30.6	(34)	
(35)	Mean Daily Temperature Range	MDBR	18.3	19.7	18.0	21.2	24.7	22.0	22.2	24.2	25.0	21.1	17.3	18.0	(35)	
(36)		5% DB	MCDBR	20.8	22.4	20.0	31.8	32.6	29.3	29.4	31.8	36.2	30.6	21.4	20.9	(36)
(37)			MCWBR	17.4	18.6	13.8	17.3	14.8	13.4	14.1	15.5	18.1	18.2	15.1	17.3	(37)
(38)			MCDBR	20.8	21.5	19.3	29.4	29.1	23.5	26.1	27.5	32.9	28.5	19.9	20.9	(38)
(39)		5% WB	MCWBR	17.5	18.3	13.8	16.9	13.6	11.8	13.5	14.5	17.0	17.6	14.5	17.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.255	0.266	0.307	0.333	0.349	0.365	0.362	0.373	0.311	0.278	0.252	0.239	(40)	
(41)		taud	2.300	2.332	2.314	2.359	2.366	2.349	2.384	2.324	2.481	2.544	2.440	2.298	(41)	
(42)		Ebn at Noon	233	270	279	285	285	280	279	268	276	266	238	217	(42)	
(43)		Edh at Noon	18	24	31	34	36	37	35	35	26	20	16	15	(43)	
(44)	All-Sky Solar Radiation	RadAvg	311	644	1131	1485	1832	1903	1956	1604	1116	643	308	230	(44)	
(45)		RadStd	35	49	88	137	131	129	99	114	118	81	35	22	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (74 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page